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RESEARCH

Dietary Information for Gut Endurance Sport Testing (DIGEST): exploring the relationship between gut health, sports nutrition knowledge and nutritional practices in elite sport

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Abstract

Gastrointestinal (GI) complaints are prevalent among athletes and can have a significant impact on performance, particularly in endurance athletes. Possessing greater nutrition knowledge and effectively applying it may play an important role in optimising an athlete's performance. Despite this, there is a paucity of research regarding GI experiences, sports nutritional knowledge (SNK) and any reciprocal relationship in elite sport. The aim of this study was to design a novel questionnaire to assess the relationship between common GI complaints, GI health, and SNK in elite athletes, sports nutritionists/dietitians and/or coaches in the United Kingdom and Ireland. The study further explored attitudes and practices regarding diet, supplementation, and training practices to gain a comprehensive understanding of potential causative factors to GI distress. This study developed and administered a novel questionnaire, the 'Dietary Information for Gut Endurance Sport Testing' (DIGEST) questionnaire to 400 elite athletes and their coaching staff from various sporting backgrounds across the United Kingdom and Ireland. Participant demographics included a balance of genders, a range of ages from 18 to 65 years, and representation from individual and team sports to ensure diverse athletic disciplines were covered. The cross-sectional questionnaire assessed GI complaints, sports nutrition knowledge (SNK), probiotic usage, dietary practices, and exercise workload. Statistical analyses, including chi-square tests and independent sample *t* tests, were employed to examine associations between these variables, providing insights into the interplay between diet, supplementation, and gastrointestinal health in elite sports settings. Among the 400 participants, 35.5% ($n = 142$) reported being diagnosed with a GI condition. The most prevalent condition was stomach cramps, affecting 27% ($n = 108$) of respondents. During physical activities, 213 athletes experienced GI complaints, with 45.5% ($n = 97$) reporting symptoms pre-exercise and 38.7% ($n = 82$) post-exercise. Specific sports showed varying levels of GI complaints, with endurance sports such as triathlons and distance running exhibiting higher rates. The study found no significant correlation between SNK scores and GI complaints ($p = 0.13$), though a substantial number of participants demonstrated suboptimal SNK scores, with athletes averaging 59% and coaches 58%. Sex-based analysis revealed no significant differences in SNK scores, although females tended to score slightly higher in certain sports. GI complaints are prevalent among elite athletes, particularly in endurance sports, with significant symptoms reported both pre- and post-exercise. Despite the lack of a significant relationship between SNK and GI complaints, improving SNK remains essential for overall athlete health and performance. Targeted nutritional education and intervention strategies are needed to manage GI complaints and enhance athlete performance.

Keywords Gastrointestinal complaints · Sports nutrition knowledge · Elite athletes · Probiotics · Dietary practices

Key points

Prevalence of GI Complaints Among Athletes: The study found that GI complaints are common among athletes, particularly in endurance sports, such as running and triathlon. These complaints often occur both pre-exercise and post-exercise, affecting performance.

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Influence of Probiotics on GI Health: While probiotic usage was common among athletes, the study showed that the relationship between probiotic usage and the reduction of GI complaints was not conclusive.

Suboptimal Sports Nutrition Knowledge: Both athletes and coaches demonstrated low scores in sports nutrition knowledge, with many not reaching the knowledge threshold that could potentially help in mitigating GI issues.

Impact of Nutrition on GI Complaints: Although the study did not find a direct correlation between SNK and the frequency of GI complaints, it highlighted the potential indirect benefits of improved nutrition knowledge, such as better dietary choices that could support gut health and possibly reduce GI symptoms.

Timing and Severity of GI Symptoms: The findings indicate that the timing of GI complaints varies, with a significant number occurring before and immediately after exercise. This variability points to the complex relationship between exercise, diet, and GI symptoms, and suggests that timing of meals and types of food consumed could be crucial in managing GI distress.

Background

Gastrointestinal (GI) complaints are a prevalent factor leading to reduced performance in athletes [1–4]. Approximately 30–70% of athletes experience GI symptoms of varying severity, including bloating, cramping, diarrhoea, and vomiting while exercising [2, 5]. These symptoms are especially prevalent among athletes engaged in endurance sports, such as distance running, triathlons, skiing, and cycling. Studies also indicate that up to 96% of ultra-distance runners experience these issues [6–8]. The aetiology of GI complaints encompasses a multifaceted interplay of physiological, nutritional, and psychological factors [5, 9]. Several mechanisms have been proposed to explain the occurrence of GI distress, including ischemia–reperfusion injury to the gut lining resulting in tight junction permeability [10], prolonged motion causing injury to the gut lining [11], and altered hormonal function involving the gut–brain axis (GBA) reducing overall GI function capacity [12]. Furthermore, a dose–response relationship exists between exercise intensity and GI complaints, indicating an increased likelihood of experiencing symptoms as exercise intensity rises [13, 14].

Efforts to mitigate GI symptoms in athletes have prompted the exploration of various preventative interventions. These include maintaining euhydration [15], carbohydrate feeding protocols [8], training the gut to tolerate and efficiently process nutrients [16], dietary fermentable oligo-di-mono-saccharide and polyols (FODMAP) modifications [14] and dietary supplements, such as probiotics, prebiotics and synbiotics [17–26]. One aspect which warrants further exploration is the concept of educational interventions, specifically targeting the enhancement of nutritional knowledge among athletes. Advanced nutritional knowledge and its effective application could significantly improve dietary decision-making, potentially reducing GI complaints and optimising performance in endurance sports. Nevertheless, there remains a paucity of research investigating any correlation between GI experiences and sports nutritional knowledge (SNK) among elite athletes.

Previous research suggests an athlete's nutritional knowledge exhibits a positive (albeit weak) correlation with improved dietary habits and subsequent performance [27–30]. To this end, several questionnaires have been developed and utilised in elite sport to establish the SNK of athletes, including 'The Nutritional Knowledge for Young and Adult Athletes (NUKYA)', 'Nutrition for Sport Knowledge Questionnaire (NSKQ)' and an abridged version of the NSKQ (the ANSKQ) [31–33]. However, there is a lack of consistency with regard to questionnaire design and factors, such as participant literacy, socioeconomic status, and education [28, 34].

While the prevalence of GI complaints in the athletic population have been extensively documented, and the influence of nutrition on GI symptoms has been acknowledged, there is still a paucity of research pertaining to the assessment of GI health knowledge levels within this population [27, 28, 35]. In addition, none of the existing questionnaires for athletes and their support staff incorporate aspects of GI health. Importantly, no studies that have investigated nutritional knowledge and GI symptoms in any population group.

Optimal nutrition is widely recognised for its role in the management of chronic diseases [36, 37]. Enhanced nutritional knowledge, coupled with positive attitudes towards healthy eating and improved dietary behaviours, are crucial factors that contribute to healthier dietary patterns and higher self-rated health. [38]. These elements underscore the need for further research and the development of comprehensive tools that include GI health considerations, to better understand and address the nutritional needs and challenges faced by athletes.

The primary objective of this study was to introduce and evaluate a novel questionnaire designed to assess GI complaints, GI health, and SNK related to GI health in elite athletes and coaching staff (sports nutritionists/dietitians and coaches). Given recent advancements in bio-sequencing technology and the heightened interest in athlete performance and probiotic usage, this study places increased emphasis on probiotic supplementation, attitudes, and practices [26, 39, 40]. A secondary aim was to explore the relationship between probiotic use and the prevalence of GI symptoms, along with other factors, such as diet, supplementation, exercise workload, and sport participation. The goal was to formulate a comprehensive understanding of

the potential causative factors contributing to GI complaints within this athlete population, providing a robust foundation for targeted nutritional interventions.

Methods

This study was conducted in two phases. Phase 1 focused on the development of the Dietary Information for Gut Endurance Sport Testing (DIGEST) questionnaire, which incorporated elements of gut health into SNK assessments. Phase 2 involved disseminating the questionnaire to athletes and coaches in elite sports across the United Kingdom and Ireland. Recruitment targeted elite athletes and coaches, leveraging existing networks within professional sports organisations, university athletic programmes, and governing bodies in both regions. Participants were approached through direct communication, institutional collaborations, and advertisements in sports performance settings.

Phase one: questionnaire development

A new SNK questionnaire, called the 'Dietary Information for Gut Endurance Sport Testing' (DIGEST) questionnaire, was developed following a scoping exercise of current nutritional knowledge screening tools used in sports nutrition professional practice and adapting previous SNK questionnaires, NSKQ [32], ANSKQ [33] and probiotics questionnaires [41, 42]. To ensure the validity of the questionnaire, a panel of experts in sports nutrition, sports medicine, and microbiology reviewed the initial draft. This panel provided feedback on the relevance, clarity, and comprehensiveness of the questions. The questionnaire included questions on demographic information, exercise information, GI complaints and frequency of occurrence, nutrition knowledge, supplement knowledge, and probiotic knowledge. The questionnaire consisted of 59 questions across 6 sub-sections: demographic information ($n = 5$), exercise information ($n = 13$), GI complaints ($n = 11$), SNK ($n = 9$), supplements ($n = 8$) and probiotic knowledge ($n = 13$). The majority of questions were multiple choice, with others requiring participant to type their answer. Demographic information consisted of questions regarding the participant's sex, age, nationality, and education level. Exercise information contained questions on the participant's sport, level they compete at (or coach), type and volume of training sessions and competition. The GI complaints section focused on the participants' GI health history, the type and frequency of GI symptoms, and the timing of these complaints in relation to exercise. Specific questions were formulated based on insights from existing literature and validated instruments. For instance, questions on GI health history and the prevalence of conditions such as IBD, GERD, reflux, constipation, and cramps were included to understand the baseline GI health of participants [41, 42] . The timing and frequency of GI symptoms were detailed to capture the exact moments these issues occur, whether before, during, or after exercise, and under what circumstances (e.g., intense or prolonged sessions). SNK and supplement questions were based on the sports nutrition sections of NSKQ [32], ANSKQ [33]. Each question in the SNK section was evaluated on a binary scoring system, where correct answers were given a score of + 1, while incorrect answers and 'unsure' responses were scored as 0. This approach allowed for a total possible score of 9, given that the SNK section consists of nine questions.

To interpret these scores, a categorical system was employed, dividing the scores into four distinct levels of knowledge: very poor, poor, good, and very good. Specifically, scores ranging from 0 to 2 were categorised as very poor knowledge, 3–4 as poor knowledge, 5–6 as good knowledge, and 7–9 as very good knowledge. According to these matrices, a score below 65% (0–5 points) was considered indicative of low or poor knowledge. To establish a framework for interpreting the participants SNK results, scoring matrices from previously published questionnaires were used [32, 43–46]. The application of these scoring matrices enables a standardised evaluation of participants' SNK, thereby facilitating the identification of knowledge gaps. For instance, a participant scoring 6 out of 9 would be classified under good knowledge, indicating a competent understanding of nutritional requirements, whereas a participant scoring 3 out of 9 would fall under poor knowledge, suggesting a need for further education and improvement in their understanding of sports nutrition. The Probiotic knowledge section of the questionnaire was adapted from previously validated instruments used to assess health professionals' understanding of probiotics [41, 42]. This section evaluated respondents' familiarity with the definition, properties, and practical application of probiotics. The questionnaire was designed to take about 15 min to complete.

A copy of the DIGEST questionnaire is included in the supplementary material to allow for transparency and facilitate future research replication.

Ethics approval and consent

This study obtained ethical approval from the Research Ethics Committee of Ulster University (SESRI-22-038-A), and all participants provided electronically signed informed consent, under the 'Declaration of Helsinki' [47].

Phase 2: recruitment

The study recruited 400 participants from the UK and Ireland, consisting of male and female elite endurance and/or high-intensity athletes, their coaches, and sports

nutritionists/dieticians in elite sport. Participants were required to meet the definition of 'elite' status, as outlined in a previous study, indicating those actively competing or striving to compete at the highest level in their respective sports [48]. They were also required to be 'healthy' within a sporting context—free from long-term injuries, diseases, or conditions that would render them non-elite. Athletes were eligible for inclusion if they were aged between 18 and 40 years and had been consistently training and competing at Olympic, international, national, or inter-county levels for a minimum of 2 years [48]. Coaches and nutritionists were eligible if aged between 18 and 65 and currently working with athletes at these performance levels Participants were excluded if they were not currently engaged in competitive activity at an elite level, had sustained a long-term injury, or had a medical condition affecting performance. Additional exclusion criteria included being under 18 years of age, failure to provide informed consent, or incomplete questionnaire submission. These inclusion and exclusion parameters were implemented to ensure a homogeneous sample representative of the target population and to enhance the internal validity of the study. Recruitment employed a multi-modal strategy to maximise reach and ensure diversity across sports. Email invitations and recruitment materials were distributed to relevant sporting organisations, including Ulster Rugby, Triathlon Ireland, Athletics Ireland, Belfast Giants Ice Hockey, Athletics Northern Ireland, and Inter-county GAA teams across the island of Ireland. Further invitations were disseminated through academic contacts at Ulster University, targeting elite athletes and coaches enrolled in university-affiliated sports programmes. In addition, social media platforms such as Facebook, LinkedIn, and Instagram were used for targeted outreach. Potential participants were identified based on their publicly available bios, indicating elite status, coaching certification, or affiliation with national teams or performance squads. A direct message was sent to eligible individuals, including information about the study, a digital consent form, and a link to the online questionnaire hosted on the REDCap survey platform (REDCap, NIH/NCATS UL1 TR000445). This approach facilitated broad recruitment while preserving participant autonomy and data confidentiality.

Data analysis

A priori power analysis was conducted using Qualtrics Online Calculator to determine the necessary sample size for detecting statistically significant differences in GI complaints among athletes [49]. Based on a two-tailed independent samples t test with a small-to-moderate effect size (Cohen's $d = 0.3$), a significance threshold of $\alpha = 0.05$, and a desired power of 80% ($\beta = 0.20$), the estimated minimum sample size was 175 participants. The final sample of 400 exceeded this requirement substantially, providing an estimated power of 99%—thereby reducing the likelihood of Type II errors. These parameters reflect accepted standards in sport nutrition research, where adequate power is critical to detect meaningful effects in performance-related outcomes [50, 51].

All statistical analyses were conducted with IBM SPSS Statistics for Windows, Version 24.0 [52]. Normality of continuous variables was assessed using the Shapiro–Wilk test. Variables that met the assumption of normality were analysed using independent samples t tests to compare group means (e.g., between athletes and coaches, or between male and female participants) on SNK scores. Where the assumption of normality was violated, non-parametric Mann–Whitney U tests were applied, particularly in the analysis of probiotic use and GI complaints. Chi-square (χ^2) tests were used to assess associations between categorical variables, including sex-based differences in the prevalence of GI conditions (e.g., IBD, cramps, bloating) and differences in GI complaints across individual vs. team sports. All relevant statistical parameters—including degrees of freedom, exact p values, and effect sizes—were reported where applicable. Cohen's d was calculated for t tests, and Cramer's V for chi-square tests to assess the magnitude of observed effects. Confidence intervals were calculated where appropriate. Descriptive statistics are presented as means ± standard deviations for continuous data and frequencies (n) and percentages (%) for categorical data. Statistical significance was set at $p \leq 0.05$.

Results

The present study successfully recruited 400 participants from the UK and Ireland, comprising 342 elite athletes and 58 coaching staff. These participants represented a diverse array of sports disciplines. A significant majority, 75% ($n = 300$), engaged in individual sports, prominently featuring distance running ($n = 110$) and triathlon ($n = 77$) (Fig. 1). The remaining 25% ($n = 100$) participated in team sports, with Gaelic football being the most common ($n = 39$) (Fig. 2). The distribution across other sports included cycling, swimming, rugby, and soccer, reflecting a broad spectrum of athletic disciplines. The demographic breakdown included 209 males (52.3%) and 191 females (47.7%), demonstrating a balanced gender representation. The age range of participants spanned from 18 to 65 years, with a median age of 28 years. The age distribution was as follows: 36% ($n = 143$) aged between 18 and 31 years, 42% ($n = 168$) aged between 32 and 50 years, and 22% ($n = 89$) aged over 50 years, illustrating a wide age range across competitive levels. Educationally, participants varied widely, with 60% ($n = 240$) holding a university degree or higher,

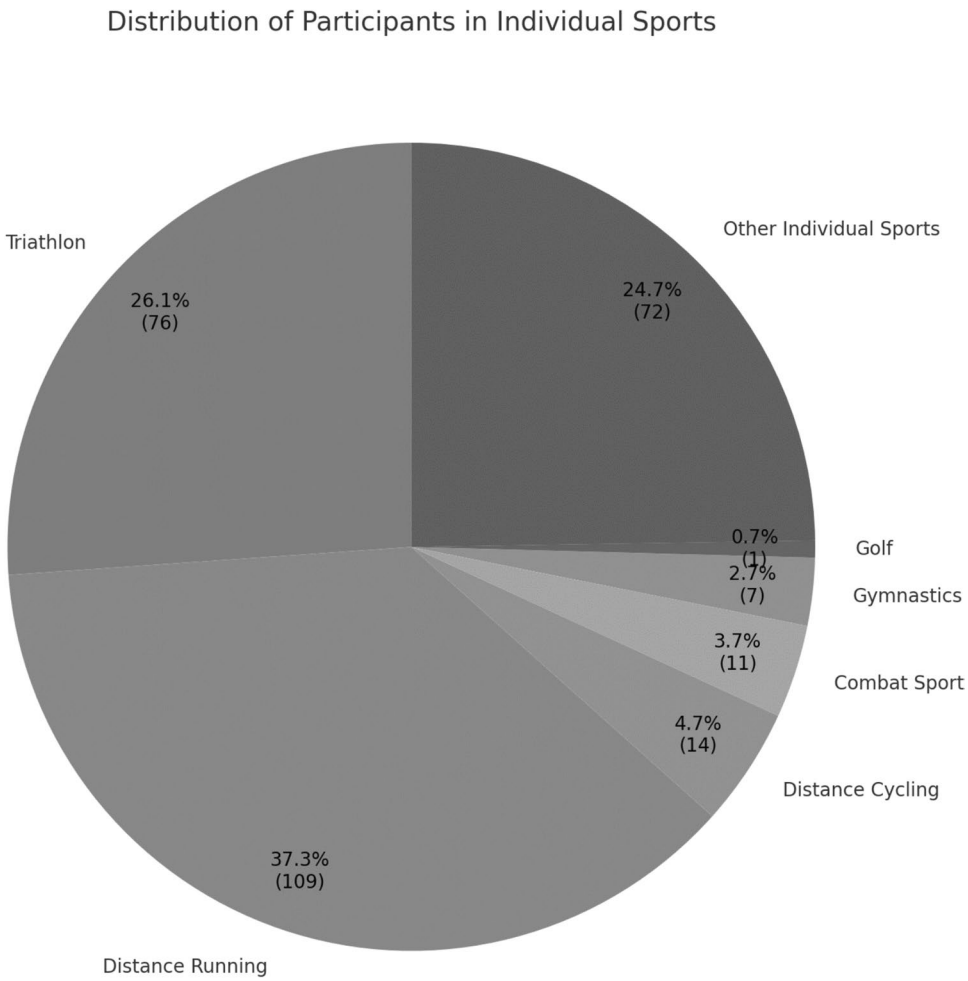

Fig. 1 HYPERLINK "sps:id::fig 1||locator::gr1||MediaObject::0" Pie chart showing distribution of participants in individual sports

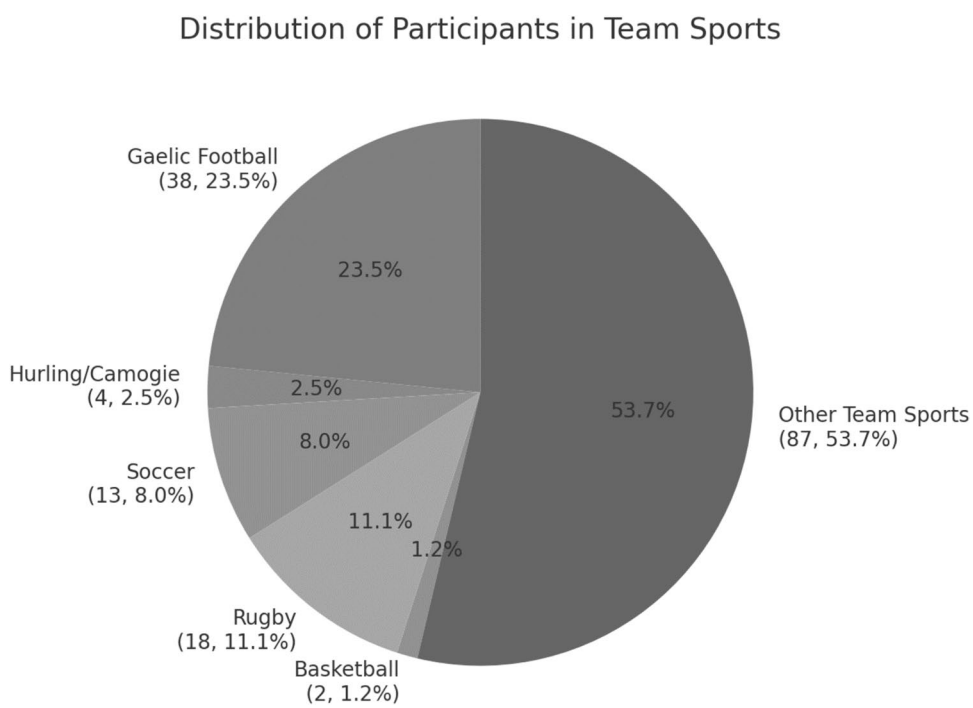

Fig. 2 Pie chart showing distribution of participants in team sports

which is relevant considering the cognitive demands of dietary self-management and sports nutrition. Approximately 20% ($n=80$) had completed secondary education, and the remaining 20% ($n=80$) held professional qualifications related to sports, such as coaching certificates or dietary certifications. The demographic distribution of participants was analysed to ensure a representative sample. The age range was assessed using a Shapiro–Wilk test, confirming normal distribution, allowing for subsequent analysis with an independent *t* test which revealed no significant age differences between sports ($p=0.157$).

GI conditions and complaints

Among the 400 study participants, 35.5% ($n=142$) reported being diagnosed with a GI condition. The most prevalent condition was stomach cramps, affecting 27% ($n=108$) of the respondents. Irritable Bowel Disease (IBD) was reported by 6.5% ($n=26$) of the participants, while gastroesophageal reflux disease (GERD) and reflux were noted by 7.3% ($n=29$) and 11.8% ($n=47$) of the participants, respectively. Constipation was present in 6.8% ($n=27$) of the participants.

During physical activities, 213 athletes experienced GI complaints, with 45.5% ($n=97$) reporting symptoms pre-exercise and 38.7% ($n=82$) post-exercise. Specific sports showed varying levels of GI complaints. Participants in hurling/camogie and basketball exhibited a 100% incidence of GI complaints, although these sports had the fewest participants ($n=4$ and $n=2$, respectively). The highest participant numbers were in triathlons, distance running, and other individual sports, with more than half of the participants in these sports reporting some form of GI complaint (58.1%, 55.3%, and 55%, respectively). Common symptoms included stomach pain and diarrhoea, with distance runners showing a higher incidence of bloating. Triathletes reported a diverse range of symptoms, including bloating, bloody diarrhoea, and stomach pain.

In triathlons, 13% ($n=10$) reported stomach pain, 13% ($n=10$) reported bloating, 4% ($n=3$) reported constipation, and 13% ($n=10$) reported diarrhoea. Among distance runners, 11% ($n=12$) reported stomach pain, 14% ($n=15$) reported bloating, 6% ($n=7$) reported constipation, and 11% ($n=12$) reported diarrhoea. Distance cyclists reported stomach pain at 36% ($n=5$), bloating at 36% ($n=5$), constipation at 14% ($n=2$), and diarrhoea at 29% ($n=4$). Among other individual sports participants, 21% ($n=15$) reported bloating, 16% ($n=12$) reported stomach pain, and 10% ($n=7$) reported constipation (Table 1).

In team sports, 33% ($n=13$) of Gaelic football participants reported no GI complaints, while 10% ($n=4$) had IBD, 3% ($n=1$) had GERD, 8% ($n=3$) experienced reflux, 3% ($n=1$) had constipation, and 18% ($n=7$) suffered from cramps. Soccer participants reported 38% ($n=5$) with no GI complaints, 15% ($n=2$) with IBD, and 8% ($n=1$) with constipation. Rugby participants reported 33% ($n=6$) with no GI complaints and 28% ($n=5$) with cramps. Participants in other team sports reported 17% ($n=15$) with no GI complaints, 1% ($n=1$) with IBD, GERD, and reflux, and 2% ($n=2$) with constipation and cramps (Table 1).

Overall, 54.7% ($n=219$) of athletes reported occasionally experiencing GI symptoms, such as stomach pain, bloating, constipation, diarrhoea, bloody diarrhoea, and nausea.

Table 1 Total participants and the percentage of GI complaints in each sport

Sport	Total participants	Stomach pain	Bloating	Constipation	Diarrhoea	Bloody diarrhoea	Nausea	None
Triathlon	77	$n=10$ (13%)	$n=10$ (13%)	$n=3$ (4%)	$n=10$ (13%)	$n=2$ (3%)	$n=6$ (8%)	$n=14$ (18%)
Distance running	110	$n=12$ (11%)	$n=15$ (14%)	$n=7$ (6%)	$n=12$ (11%)	$n=2$ (2%)	$n=9$ (8.18%)	$n=18$ (16%)
Distance cycling	14	$n=5$ (36%)	$n=5$ (36%)	$n=2$ (14%)	$n=4$ (29%)	$n=1$ (7%)	$n=2$ (14%)	$n=7$ (50%)
Combat sport	11	$n=3$ (27%)	$n=3$ (27%)	$n=1$ (9%)	$n=2$ (18%)	$n=0$ (0%)	$n=2$ (18%)	$n=5$ (45%)
Gymnastics	8	$n=2$ (25%)	$n=4$ (50%)	$n=2$ (25%)	$n=3$ (38%)	$n=0$ (0%)	$n=1$ (13%)	$n=7$ (88%)
Golf	2	$n=0$ (0%)	$n=1$ (50%)	$n=0$ (0%)	$n=0$ (0%)	$n=0$ (0%)	$n=0$ (0%)	$n=1$ (50%)
Other individual sports	73	$n=12$ (16%)	$n=15$ (21%)	$n=7$ (10%)	$n=9$ (12%)	$n=2$ (3%)	$n=6$ (8%)	$n=20$ (27%)
Gaelic Football	39	$n=2$ (5%)	$n=3$ (8%)	$n=1$ (3%)	$n=2$ (5%)	$n=0$ (0%)	$n=1$ (3%)	$n=5$ (13%)
Hurling/camogie	4	$n=1$ (25%)	$n=1$ (25%)	$n=0$ (0%)	$n=0$ (0%)	$n=0$ (0%)	$n=0$ (0%)	$n=2$ (50%)
Soccer	13	$n=4$ (31%)	$n=3$ (23%)	$n=1$ (8%)	$n=3$ (23%)	$n=0$ (0%)	$n=2$ (15%)	$n=5$ (38%)
Rugby	18	$n=2$ (11%)	$n=2$ (11%)	$n=1$ (6%)	$n=2$ (11%)	$n=0$ (0%)	$n=2$ (11%)	$n=6$ (33%)
Basketball	2	$n=0$	$n=1$ (50%)	$n=0$	$n=0$	$n=0$	$n=0$	$n=1$ (50%)
Other team sports	87	$n=10$ (11%)	$n=12$ (14%)	$n=5$ (6%)	$n=8$ (9%)	$n=1$ (1%)	$n=4$ (5%)	$n=15$ (17%)
Total (participants could play both individual and team sport	458							

Symptoms were more frequently reported during longer ($n=32$, 14.8%) and more intense ($n=38$, 17.7%) exercise sessions. A weak positive correlation was observed between exercise duration and both bloating (correlation coefficient of 0.15) and diarrhoea (0.18), as well as between exercise intensity and both stomach pain (0.11) and diarrhoea (0.24). In addition, 31.75% ($n=127$) of participants sought professional advice for digestive issues, with 35.5% ($n=142$) diagnosed with conditions, such as IBS, GERD, reflux, constipation, and cramps.

A chi-square test for independence indicated a statistically significant difference in the frequency of GI complaints between athletes in individual vs. team sports ($\chi^2(1, N=400)=4.17$, $p=0.041$, $\varphi=0.10$), with those in individual sports reporting a higher overall number of complaints ($n=161$) compared to their team sport counterparts ($n=50$). Although the effect size was small, these findings suggest that sport type may influence the prevalence of GI symptoms. Table 1 provides a detailed breakdown of GI symptoms reported across each sport. Participants could select multiple symptoms, meaning total counts may exceed the number of participants per sport.

Sex-based differences in gastrointestinal complaints

A comparative analysis was conducted to explore sex-based differences in the prevalence of GI complaints among athletes (Table 2). The 400 participants consisted of 209 males (52.3%) and 191 females (47.7%). Among the male participants, 67.0% ($n=140$) reported having no GI condition. For those with GI conditions, 7.2% ($n=15$) reported IBD, 7.2% ($n=15$) reported GERD, 13.4% ($n=28$) reported reflux, and 8.1% ($n=17$) reported constipation. In addition, 11.5% of males ($n=24$) experienced stomach cramps. In comparison, 54.5% of female participants ($n=104$) reported having no GI condition. Among those with GI conditions, 8.4% ($n=16$) reported IBD, 4.2% ($n=8$) reported GERD, 11.5% ($n=22$) reported reflux, 8.9% ($n=17$) reported constipation, and 14.1% ($n=27$) experienced stomach cramps.

Table 2 Sex differences in diagnosed GI conditions among the study participants

Condition	Male count ($n=209$)	Male Percentage	Female count ($n=191$)	Female percentage
No GI issue	140	67.0%	104	54.5%
IBD	15	7.2%	16	8.4%
GERD	15	7.2%	8	4.2%
Reflux	28	13.4%	22	11.5%
Constipation	17	8.1%	17	8.9%
Cramps	24	11.5%	27	14.1%

GI complaints during exercise were also examined (Table 3). Among male athletes, 33.0% ($n=69$) reported such complaints. Pre-exercise symptoms were reported by 7.2% ($n=15$), while post-exercise symptoms were noted by 16.7% ($n=35$). The most common symptoms among males included stomach pain (21.5%, $n=45$), bloating (18.2%, $n=38$), and diarrhoea (13.4%, $n=28$). In contrast, 45.5% of female athletes ($n=87$) reported GI complaints during exercise, with 6.3% ($n=12$) experiencing symptoms pre-exercise and 16.8% ($n=32$) post-exercise. The most frequent symptoms for females were stomach pain (26.2%, $n=50$), bloating (22.5%, $n=43$), and diarrhoea (16.2%, $n=31$).

In triathlons, males reported higher incidences of IBD (9.0%) and reflux (11.7%) compared to females (7.8% and 9.1%, respectively). Conversely, females reported more cases of GERD (10.4%) and constipation (6.5%) than males (8.0% and 5.2%, respectively). Among distance runners ($n=110$), males experienced higher rates of reflux (10.9%) and cramps (13.6%) compared to females (8.1% and 10.0%, respectively), while females had higher instances of bloating (15.5%) and constipation (7.3%) compared to males (13.1% and 6.4%, respectively).

For combat sports participants ($n=11$), males had a higher prevalence of cramps (18.2%) compared to females (11.1%), whereas females reported more IBD (11.1%) and GERD (9.1%) than males (9.1% and 0.0%, respectively). In distance cycling ($n=14$), males showed higher rates of cramps (7.1%) and bloating (21.4%) compared to females (5.7% and 14.3%, respectively), while females reported more IBD (7.1%) compared to males (6.4%).

Chi-square tests were conducted to determine if there were significant differences in the prevalence of GI complaints between male and female athletes. The results indicated that the differences in the prevalence of IBD ($\chi^2=4.12$, $p=0.042$), stomach cramps ($\chi^2=5.76$, $p=0.016$), and bloating ($\chi^2=4.96$, $p=0.026$) were statistically significant, with female athletes reporting higher incidences of these conditions. However, the differences in the prevalence of GERD ($\chi^2=1.32$, $p=0.251$), reflux ($\chi^2=2.11$, $p=0.146$),

Table 3 Sex differences in GI complaints during physical activity

Complaint	Male Count ($n=209$)	Male Percentage	Female Count ($n=191$)	Female Percentage
Stomach Pain	45	21.5%	50	26.2%
Bloating	38	18.2%	43	22.5%
Constipation	14	6.7%	15	7.9%
Diarrhea	28	13.4%	31	16.2%
Bloody Diarrhea	5	2.4%	6	3.1%
Nausea	18	8.6%	20	10.5%
None	140	67.0%	104	54.5%

and constipation ($\chi^2 = 1.09$, $p = 0.296$) were not statistically significant.

Sports nutrition knowledge

Results presented as percentage mean ± standard deviation, unless otherwise indicated. Differences between athlete and coaches, and gender based on independent samples *t* test ($p < 0.05$).

In accordance with the methodology described, we utilised scoring matrices from previously published studies to categorise SNK scores into four distinct classifications: very poor, poor, good, and very good [32, 43–46]. The 400 participants consisted of 342 athletes and 58 coaches, of these, 242 participants scored below 65% in the SNK assessment, falling into the 'poor knowledge' category as per the scoring matrices (Table 4).

Athletes had a mean score of 4.98 (SD = 1.56), while coaches scored slightly higher with a mean of 5.22 (SD = 1.61), indicating no significant difference in nutrition knowledge between athletes and coaches ($p \geq 0.05$). The highest average SNK scores were noted in basketball, at 88.89, although there was a limited sample size of just two participants. Conversely, combat sports and other team sports exhibited the lowest average scores, 51.51 and 50.85, respectively. No significant difference was found when comparing SNK of athletes and coaches or between genders ($p > 0.05$). However, a consistent trend observed across the studied sports was the higher SNK scores achieved by coaches compared to athletes, especially pronounced in individual sports, with an overall average difference of approximately 14.24%. Sex-based analysis of the data revealed higher SNK scores among female athletes in several sports, including rugby, Gaelic football, and hurling, compared to their male counterparts (overall average approximately 20.28%).

The impact of SNK on GI complaints was analysed using an independent *t* test, which found no significant differences in GI complaint rates between athletes with high and low SNK scores ($p = 0.109$).

Sports nutrition knowledge and GI complaints

Among the participants, 213 reported experiencing GI complaints and had low SNK scores (53.3%), while 187 with high SNK scores (46.7%) also reported such complaints. Statistical analysis revealed no significant difference in the incidence of GI complaints between the two groups. This suggests that SNK scores do not markedly influence the frequency or severity of GI complaints.

Of the 400 participants, 186 (46.5%) reported that they currently purchase and consume probiotics, while 204 (51.0%) reported that they do not. Among non-consumers, the most common reason for non-use was uncertainty about the benefits of probiotics (54.9%, $n = 112$), followed by cost concerns (23.5%, $n = 48$), lack of awareness (13.2%, $n = 27$), and difficulty accessing reliable products (8.3%, $n = 17$). These results highlight both informational and practical barriers influencing probiotic use among elite athletes.

The relationship between probiotic use and GI complaints is presented in Table 5. A higher proportion of non-users (56.4%, $n = 115$) reported no GI complaints, compared to probiotic users (37.6%, $n = 70$). Conversely, 62.4% of

Table 4 Comparison of SNK scores between type of sport, athletes and coaches, gender and overall score for sport

Sport	Total participants	Athlete SNK score %	Coach SNK score%	Male SNK score %	Female SNK SCORE %	SNK score %
Individual sports:						
Triathlon	77	54.26 ± 20.8	63.5 ± 12.9	57.73 ± 19.6	51.4 ± 19.5	55 ± 19.52
Distance running	110	54.72 ± 16.64	56.95 ± 10.3	58.21 ± 15.6	56.31 ± 16.14	54.93 ± 16.19
Distance cycling	14	61.48 ± 11.38		64.2 ± 10.18	57.4 ± 11.86	61.48 ± 11.38
Combat sport	11	51.85 ± 18.15	50 ± 5.56	57.14 ± 12.5	37.04 ± 18.89	51.51 ± 16.6
Other	83	54.06 ± 15.94	69.44 ± 9.21	55.56 ± 16.44	56.16 ± 15.7	55.7 ± 16.08
Team sports:						
Gaelic Football	39	50.84 ± 16.19	65.08 ± 22.56	51.51 ± 17.26	63.89 ± 26.5	53.33 ± 18.29
Hurling or Camogie	4	40.74 ± 18.89	8	44.45 ± 22.26	61.11 ± 27.78	52.78 ± 26.5
Soccer	13	52.78 ± 13.32	66.67 ± 25.34	55.56 ± 18.15	66.67 ± 22.22	55.56 ± 17.08
Rugby	18	59.83 ± 14.8	46.66 ± 10.89	54.32 ± 13.3	66.67 ± 15.7	56.17 ± 15.03
Basketball	2		88.89		88.89	88.89
Other	87	50.55 ± 13.82	51.85 ± 19.95	47 ± 12.45	54.7 ± 17.14	50.85 ± 15.46
	458					

Table 5 Probiotic use and GI complaints among participants

Probiotic use	Participants	No GI complaints (*n*, %)	GI complaints (*n*, %)
Consumers	186	70 (37.6%)	116 (62.4%)
Non-consumers	204	115 (56.4%)	89 (43.6%)

probiotic users ($n = 116$) reported experiencing GI complaints, compared to 43.6% ($n = 89$) of non-users.

To statistically assess the difference in GI complaint frequency between users and non-users, a Mann–Whitney *U* test was conducted due to the non-parametric distribution of GI symptom frequency. Results indicated that the difference in GI complaints between probiotic users and non-users was not statistically significant ($U = 16{,}639$, $p = 0.074$), suggesting no conclusive evidence that probiotic use reduced or increased the frequency of GI symptoms in this population.

However, further exploratory chi-square analyses revealed significant associations between probiotic use and the presence of specific conditions. Athletes reporting bloating were significantly more likely to use probiotics ($\chi^2(1, N = 400) = 5.48$, $p = 0.019$, $\varphi = 0.12$), as were those reporting IBD ($\chi^2(1, N = 400) = 4.21$, $p = 0.040$, $\varphi = 0.10$). These small effect sizes suggest that athletes experiencing certain GI symptoms may be more inclined to self-manage with probiotics, even though the overall relationship between use and symptom burden remains inconclusive.

Probiotic use varied considerably by sport, as summarised in Table 6. Higher usage was reported in distance running (60.0%) and triathlon (64.9%), which also had high overall GI complaint rates. In contrast, team sports showed markedly lower probiotic use (27.6%), despite moderate complaint prevalence. Among distance runners and triathletes, GI complaints were reported by both probiotic users (62.1% and 68.0%, respectively) and non-users (50.0% and 53.3%, respectively), indicating that the presence of symptoms persisted regardless of probiotic use.

Discussion

The pathophysiology of GI distress in athletes is highly heterogeneous, preventing the identification of a single causal factor. Although various hypotheses have been proposed, the precise mechanisms remain elusive. Nevertheless, it is widely acknowledged that nutrition can play a significant role, both beneficial and detrimental, in influencing GI symptoms in this context [2, 11, 16, 53–56]. Factors such as meal composition and hydration status can dramatically influence GI function during exercise [34]. While previous research indicates that SNK can shape athletes' dietary habits, there is a paucity of research exploring any direct influence SNK may have on GI complaints. Therefore, the primary aim of this study was to introduce and evaluate a novel questionnaire for assessing prevalent GI complaints, GI health, and SNK in elite athletes and coaches.

GI complaints

Our results add to the body of evidence indicating GI complaints are prevalent among elite athletes with a total of 53% (213 complaints) reported during training and/or competition. In addition, 142 participants disclosed a diagnosed GI condition relating to stomach cramps, IBD, GERD and reflux. Endurance sports, predominantly individual-based like running and cycling, have been well-documented for their association with GI complaints. Studies have consistently shown a high incidence of GI symptoms among endurance runners, with reports suggesting up to 90% of athletes experiencing severe symptoms during their activities [57, 58]. Similarly, a significant proportion of other endurance athletes, including cyclists and triathletes, are affected, with the occurrence ranging from 30 to 90% [56]. In one study, 95–96% of ultramarathon runners in a 161-km race

Table 6 Probiotic use by sport and associated GI complaints

Sport	Probiotic users (*n*, %)	GI complaints in users (*n*, %)	GI complaints in non-users (*n*, %)
Triathlon	50 (64.9%)	34 (68.0%)	16 (53.3%)
Distance Running	66 (60.0%)	41 (62.1%)	25 (50.0%)
Distance cycling	7 (50.0%)	4 (57.1%)	3 (42.9%)
Combat sports	3 (27.3%)	2 (66.7%)	1 (33.3%)
Other individual	36 (43.4%)	21 (58.3%)	15 (41.7%)
Team sports	24 (27.6%)	14 (58.3%)	10 (41.7%)

experienced some form of GI symptoms [57]. Our observed incidence rate of 53% not only falls within this documented range but also serves to reinforce the established connection between endurance sports and GI distress.

Endurance athletes face a heightened risk of GI issues due to several factors. The prolonged physical exertion inherent in these sports leads to a significant diversion of blood flow away from the digestive tract—splanchnic perfusion can drop by up to 80% during intense exercise [59]. This intestinal ischemia contributes to symptoms, such as cramping, nausea, and even mucosal damage. In addition, the mechanical jostling experienced during high-impact activities (such as running) likely exacerbates GI disturbances. Repetitive impact can traumatise the gut; for instance, running has been associated with more GI complaints (and even GI bleeding) than cycling, largely due to the gut's repetitive impact stress combined with reduced blood flow [2]. Our data, showing endurance athletes (predominantly in individual sports, such as running and cycling) reporting more frequent GI issues, align with these physiological and mechanical mechanisms described in the literature.

Sex-based differences in gastrointestinal complaints

The study revealed significant sex-based differences in the prevalence and type of GI complaints among athletes. Contrary to some earlier assumptions that male athletes might be more susceptible to exercise-induced GI symptoms (owing to generally higher training intensities and greater physiological stress) [60], our results showed a higher overall incidence of GI complaints among female participants—with 45.5% of females reporting issues, compared to 33.0% of males. Interestingly, other recent research has likewise observed that female endurance athletes tend to report more GI symptoms than their male counterparts [60]. In our cohort, female athletes not only had a greater overall frequency of GI problems, but also reported higher incidences of specific diagnosed GI conditions. Specifically, among females in this study, 8.4% reported IBD, 4.2% GERD, 11.5% reflux, and 8.9% constipation, with 14.1% experiencing recurrent stomach cramps. In comparison, male athletes reported IBD (7.2%), GERD (7.2%), reflux (13.4%), constipation (8.1%), and stomach cramps (11.5%).

GI complaints during exercise also demonstrated sex-based patterns. Female athletes more frequently experienced symptoms such as stomach pain (26.2% of females vs. 21.5% of males), bloating (22.5% vs. 18.2%), and diarrhoea (16.2% vs. 13.4%) during or immediately following exercise. These findings suggest that female athletes may be more prone to certain GI issues during physical activity. One potential explanation is the influence of hormonal fluctuations—particularly those related to the menstrual cycle—on GI function. It is known that gut motility and sensitivity can be affected by female sex hormones; for example, female athletes have a higher prevalence of irritable bowel syndrome and often report that GI symptoms worsen around menstruation [61]. Slower GI transit in the luteal phase and other hormonal effects can lead to symptoms such as bloating, cramps, and constipation in women. Our data did not pinpoint menstrual cycle phase, but the higher incidence of conditions such as stomach cramps, IBD, and bloating among females is consistent with the notion that hormonal factors may contribute to GI vulnerability in female athletes.

Notably, chi-square analyses in this study confirmed significant sex-based differences in the prevalence of certain GI conditions: IBD ($\chi^2 = 4.12$, $p = 0.042$), stomach cramps ($\chi^2 = 5.76$, $p = 0.016$), and bloating ($\chi^2 = 4.96$, $p = 0.026$) were all more common in females. In contrast, no significant differences were found between sexes for GERD, reflux, or constipation ($p > 0.14$ in each case). These results highlight the need for sex-specific strategies in managing and preventing GI complaints. The higher prevalence of issues such as stomach cramps, IBD, and bloating in female athletes suggests that interventions (nutritional adjustments, medical management, etc.) might need to account for female-specific factors such as hormonal cycles. Conversely, while male athletes showed a lower overall rate of GI complaints, certain issues (e.g., reflux in some sports) still affected them substantially, indicating that both sexes benefit from targeted GI management strategies. Overall, understanding these differences is crucial for developing effective, individualised approaches to enhance all athletes' performance and well-being.

Timing of GI symptoms and nutritional impact

Most participants experienced GI complaints before starting their exercise, as indicated by 45.5% of respondents, with the next most common timing being immediately post-exercise (38.7%). It should be noted that the questionnaire allowed for multiple responses regarding the timing of symptoms, reflecting the multifaceted nature of GI distress in the context of physical activity. This pattern challenges the popular belief of a simple temporal association between exercise and the onset of symptoms [11]. The predominance of GI symptoms prior to exercise initiation underscores a more complex relationship between physical activity and GI distress than previously understood.

The increased occurrence of symptoms such as bloating and diarrhoea in longer and more intense exercise sessions might reflect the impact of prolonged physical strain and dehydration on GI function and inflammatory pathways [62, 63]. The high prevalence of pre-exercise GI complaints observed in this study suggests other factors influence symptoms, and perhaps further exploration of the relationship between GI distress and the gut–brain axis is warranted.

Understanding the timing of GI complaints during exercise is crucial for developing targeted interventions and optimising athletes' performance and well-being.

Pre-exercise nutrition can heavily influence the occurrence and severity of GI complaints. Consuming large meals or certain types of foods close to exercise can increase the likelihood of symptoms, such as reflux, bloating, and cramps [16]. Nutritional strategies to mitigate these issues include consuming easily digestible foods and avoiding high-fat and high-fibre foods pre-exercise [2, 16, 64, 65]. Symptoms occurring immediately post-exercise were also common in our cohort. Over one-third (38.7%) of athletes reported GI issues after finishing exercise, highlighting that the aftermath of intense activity is a vulnerable period for GI discomfort. Prolonged, strenuous exercise can leave the gut in a compromised state—reduced blood flow, mild ischemia, and accumulated metabolic stress can persist even after exercise stops, leading to issues such as nausea, bloating, or diarrhoea during recovery. The increased occurrence of symptoms such as bloating and diarrhoea following longer, or more intense exercise sessions likely reflects the impact of prolonged physical strain and dehydration on GI function and inflammatory pathways. Dehydration, in particular, is known to exacerbate GI distress by further impairing gastrointestinal perfusion and function [54, 56, 66]. Even a moderate fluid deficit can concentrate the gut contents and slow gastric emptying, increasing the likelihood of discomfort. One study demonstrated that dehydration resulting from inadequate fluid intake was associated with higher incidence of GI complaints during a triathlon [2].

Sports nutrition knowledge

To establish a benchmark for interpreting SNK scores and guide future research, several studies employed scoring matrices categorised into four categories: very poor, poor, good, and very good [32, 43–46]. Among the participants in this study, 242 scored below 65% in the SNK section, indicating inadequate or poor SNK knowledge, with a mean SNK percentage of 55%. These results are consistent with previous studies that have investigated SNK. For instance, a previous study found that Division 1 college athletes had a mean score of 45.7% [67], similarly, two other studies investigating Gaelic Footballers' nutritional knowledge reported mean scores of 44.3% and 54% [68, 69]. Our data also highlighted interesting variations among different sports. Basketball players, albeit in a small sample size, showed the highest average SNK score of 88.89. In contrast, athletes in combat sports and team sports such as rugby and Gaelic football scored lower, with averages of 51.51 and 50.85, respectively. This variation suggests that SNK may be influenced by the nature of the sport and the emphasis placed on nutrition in training. In addition, our sex-based analysis of SNK revealed that female athletes in several sports scored slightly higher on nutrition knowledge than their male counterparts. This trend has been noted in some other studies as well; for example, higher general nutrition knowledge has sometimes been observed among female athletes or female students compared to males [28]. The reasons are not certain, but it could be that female athletes are more likely to seek nutrition information, or perhaps they receive more guidance related to energy availability and diet.

Moreover, the coaches in this study had a higher mean SNK score than the athletes (particularly in individual sports), which is an expected but important finding. Coaches (and support staff such as nutritionists where available) often receive more formal education or exposure to sports nutrition information. In our sample, this translated to coaches outperforming athletes on the knowledge assessment. This disparity likely reflects a knowledge transfer gap, i.e., even when coaches hold sound nutrition knowledge, this information is not being translated into improved athlete nutritional knowledge. These findings suggest an opportunity for improvement by refining communication and education strategies, enabling coaches to effectively convey nutrition knowledge to athletes with the support of qualified sport and exercise nutritionists.

GI complaints and association with SNK score

In terms of the potential association between SNK and GI complaints among the participants, we found that 242 individuals with low SNK scores reported experiencing GI complaints, while 158 with high SNK scores also reported such issues, indicating no significant difference between SNK score and the occurrence of GI complaints. These results suggest that sports nutrition knowledge alone may not have a direct association with the occurrence of GI complaints in elite athletes. This observation is consistent with the broader research landscape, which points to multiple factors influencing GI complaints in athletes, such as physiological stress, psychological factors, and lifestyle choices [12, 16, 64, 65, 70]. However, it is important to consider the potential indirect benefits of improved SNK [28, 29, 71, 72]. For example, a study by Nascimento and colleagues [71] demonstrated that enhanced nutrition knowledge led to better food choices and dietary habits among athletes, which subsequently improved overall health markers. Similarly, Spronk and colleagues [28] found a positive correlation between high levels of SNK and improved dietary habits, suggesting a beneficial impact on athletes' health and performance. While our research did not find a direct link between SNK and GI complaints, this study differs from existing research by specifically focusing on the direct correlation between SNK and GI complaints, adding a unique perspective to the existing body of knowledge. Improved SNK, encompassing

elements of GI health as highlighted in our questionnaire, could indirectly influence GI health through more informed dietary and lifestyle choices. For instance, better nutrition knowledge may lead athletes to adopt diets that support gut health, potentially mitigating some GI complaints. This indirect pathway suggests that while SNK may not directly reduce the incidence of GI complaints, it plays a crucial role in shaping behaviours and choices that contribute to overall GI and athletic health. However, exploring the specific mechanisms through which enhanced SNK might lead to improvements in GI health and determining the extent of these potential benefits was outside the scope of our current study.

Individual vs. *team sport*

Our analysis revealed that athletes in individual sports, especially in endurance activities, such as triathlons and distance running, reported more GI complaints, such as diarrhoea and bloating, compared to those in team sports. This aligns with the current literature suggesting a higher susceptibility to GI symptoms among individual sports athletes [11, 65, 73, 74]. The mechanism behind this increased vulnerability may be a result of the prolonged physical exertion and associated physiological changes in these sports, such as altered gut permeability and reduced blood flow to the GI tract [11]. In marathon runners, for example, severe GI distress has been observed in a minority of competitors (around 4% in one study), whereas in ultra-endurance events like Ironman triathlons, rates of severe GI issues can reach ~ 32% of participants [3, 75]. By contrast, team sports (such as soccer, rugby, or basketball) typically involve intermittent activity—bursts of high intensity interspersed with periods of lower intensity or rest—and generally last shorter durations than endurance races [8, 76]. These differences likely explain why team sport athletes in this study experienced relatively fewer GI complaints. The stop-and-go nature may allow periodic restoration of splanchnic blood flow and reduces the prolonged jostling of the gut, thereby lessening GI distress during play. Our findings suggest that while team sport athletes are not immune to GI issues, they may face different triggers or lower overall risk compared to endurance athletes. For practitioners, this means GI management strategies might need to be sport specific.

Probiotics

This study represents an initial exploration of probiotic knowledge among elite athletes and their coaches, with no previous data available for comparison. However, findings from Soni and colleagues [41] on health professionals' attitudes and practices regarding probiotic use revealed that although participants were aware of the term "probiotic," only 54.7% knew about their reported benefits. Similarly, this study found many (69.2%) of the elite athletes and coaches expressed uncertainty regarding the potential benefits of probiotics, choosing the option 'I'm not sure how probiotics would benefit me'. Interestingly, while a considerable number of participants reported using probiotics within the current study, GI symptoms continued to be a common concern across various sports disciplines. This suggests that athletes might be using probiotics as a self-directed approach to managing specific GI issues, such as stomach pain, bloating, and IBD [19, 20, 22, 62, 77]. This aligns with existing literature, which indicates the frequent use of probiotics may alleviate symptoms of GI disorders [78, 79]. For instance, Cox et al. reported that endurance athletes who took a specific probiotic (*Lactobacillus fermentum*) had fewer days of respiratory illness and hinted at immune and gut benefits, illustrating systemic advantages of maintaining gut health [78]. More directly, a study by Pugh and colleagues demonstrated that 4 weeks of probiotic supplementation significantly reduced GI symptoms during a marathon race [25]. Similarly, Schreiber et al. [79] conducted a trial with elite cyclists and found that those taking a probiotic experienced fewer GI complaints during training, as well as reduced inflammatory markers and improved immune profiles [79]It is important to note that the benefits of probiotics are strain-specific and context-dependent, the survey primarily focused on whether participants were taking probiotics, without delving into the nuances of their usage, such as the duration, type, and perceived efficacy of these probiotics. Emerging research, however, points towards the potential benefits of probiotics in alleviating GI issues, particularly in endurance athletes. Studies suggest that probiotics may enhance gut health and contribute to maintaining gut barrier integrity, potentially reducing the occurrence or severity of GI symptoms [21, 25, 62, 77, 79].

Limitations

This study is not without limitations, first, the online administration of the questionnaire, while facilitating broad participation, introduces the possibility of non-authentic responses. However, this concern is unlikely given the low SNK scores, making it less likely that participants engaged in deliberate cheating. Furthermore, the inclusion of coaches in the study was relatively low compared to the athlete population. This discrepancy in sample size may affect the generalisability of the findings, and therefore, consideration must be taken when interpreting the results related to coaches' perspectives. One of the key limitations of this study lies in the nature of the questionnaire used. The survey predominantly focused on whether participants were consuming probiotics without delving deeper into the specifics of their use, such as duration, type, dosage, and most crucially, the athletes'

subjective assessment of the effectiveness of probiotics on their symptoms and overall performance. This gap in data collection limits the scope of our understanding of the true impact of probiotics in managing GI health among athletes. In addition, certain important aspects of athletes' dietary practices, such as meal timing, hydration status, and supplement intake, were not comprehensively examined in this study. The extent to which these factors contribute to the occurrence of GI complaints remains unclear and should be further investigated in future research.

Conclusion

Despite advancements in sports nutrition, the findings in this study confirm that GI complaints are still prevalent among athletes. Furthermore, the study reveals that both athletes and their coaches possess suboptimal sports nutrition knowledge. The development of the DIGEST questionnaire, tailored specifically for athletes and coaches, offers potential as a tool for assessing gut health, GI complaints, and SNK during training and competition. By implementing the DIGEST questionnaire, nutritionists, dieticians, and coaches can identify athletes who are experiencing GI complaints and subsequently provide targeted interventions to effectively manage their symptoms. The findings underscore the importance of addressing nutrition education, optimising gut health, and considering individualised strategies to manage GI distress in athletes.

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Declarations

Conflict of interests The authors declare no competing interests.

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References

1. Jeukendrup AE, Vet-Joop K, Sturk A, Stegen JHJC, Senden J, Saris WHM et al (2000) Relationship between gastro-intestinal complaints and endotoxaemia, cytokine release and the acute-phase reaction during and after a long-distance triathlon in highly trained men. Clin Sci. https://doi.org/10.1042/cs0980047
2. De Oliveira EP, Burini RC, Jeukendrup A (2014) Gastrointestinal complaints during exercise: prevalence, etiology, and nutritional recommendations. Sports Med. https://doi.org/10.1007/s40279-014-0153-2
3. Hoogervorst D, van der Burg N, Versteegen JJ, Lambrechtse KJ, Redegeld MI, Cornelissen LAJ et al (2019) Gastrointestinal complaints and correlations with self-reported macronutrient intake in independent groups of (Ultra)marathon runners competing at different distances. Sports. https://doi.org/10.3390/sports7060140
4. Marttinen M, Ala-Jaakkola R, Laitila A, Lehtinen MJ (2020) Gut microbiota, probiotics and physical performance in athletes and physically active individuals. Nutrients. https://doi.org/10.3390/nu12102936
5. Waterman JJ, Kapur R (2012) Upper gastrointestinal issues in athletes. Curr Sports Med Rep. https://doi.org/10.1249/JSR.0b013e318249c311
6. van Wijck K, Lenaerts K, Grootjans J, Wijnands KAP, Poeze M, van Loon LJC et al (2012) Physiology and pathophysiology of splanchnic hypoperfusion and intestinal injury during exercise: strategies for evaluation and prevention. Am J Physiol Gastrointest Liver Physiol. https://doi.org/10.1152/ajpgi.00066.2012
7. Schwellnus MP, Wright J. Gastrointestinal system and exercise: a clinical approach to gastrointestinal problems encountered in athletes. In: *Olympic Textbook of Medicine in Sport*. 2009; https://doi.org/10.1002/9781444300635.ch11.
8. Costa RJS, Snipe RMJ, Kitic CM, Gibson PR (2017) Systematic review: exercise-induced gastrointestinal syndrome—implications for health and intestinal disease. Aliment Pharmacol Ther. https://doi.org/10.1111/apt.14157
9. De Oliveira EP, Burini RC (2009) The impact of physical exercise on the gastrointestinal tract. Curr Opin Clin Nutr Metab Care. https://doi.org/10.1097/MCO.0b013e32832e6776
10. Ter Steege RWF, Kolkman JJ, Huisman AB, Geelkerken RH. Gastrointestinal ischaemia during physical exertion as a cause of gastrointestinal symptoms. *Ned Tijdschr Geneeskd* 2008; 152.
11. Smith KA, Pugh JN, Duca FA, Close GL, Ormsbee MJ (2021) Gastrointestinal pathophysiology during endurance exercise: endocrine, microbiome, and nutritional influences. Eur J Appl Physiol. https://doi.org/10.1007/s00421-021-04737-x

12. Clark A, Mach N (2016) Exercise-induced stress behavior, gut-microbiota-brain axis and diet: A systematic review for athletes. J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-016-0155-6
13. Costa RJS, Snipe R, Camões-Costa V, Scheer V, Murray A (2016) The impact of gastrointestinal symptoms and dermatological injuries on nutritional intake and hydration status during ultramarathon events. Sports Med Open. https://doi.org/10.1186/s40798-015-0041-9
14. Costa RJS, Gaskell SK, McCubbin AJ, Snipe RMJ (2020) Exertional-heat stress-associated gastrointestinal perturbations during Olympic sports: management strategies for athletes preparing and competing in the 2020 Tokyo Olympic Games. Temperature. https://doi.org/10.1080/23328940.2019.1597676
15. Costa RJS, Hoffman MD, Stellingwerff T (2019) Considerations for ultra-endurance activities: part 1-nutrition. Res Sports Med. https://doi.org/10.1080/15438627.2018.1502188
16. Jeukendrup AE (2017) Training the gut for athletes. Sports Med. https://doi.org/10.1007/s40279-017-0690-6
17. Calero CDQ, Rincón EO, Marqueta PM (2020) Probiotics, prebiotics and synbiotics: useful for athletes and active individuals? A systematic review. Benef Microbes. https://doi.org/10.3920/BM2019.0076
18. Miles MP (2020) Probiotics and gut health in athletes. Curr Nutr Rep. https://doi.org/10.1007/s13668-020-00316-2
19. Roberts JD, Suckling CA, Peedle GY, Murphy JA, Dawkins TG, Roberts MG (2016) An exploratory investigation of endotoxin levels in novice long distance triathletes, and the effects of a multi-strain probiotic/prebiotic, antioxidant intervention. Nutrients. https://doi.org/10.3390/nu8110733
20. Lamprecht M, Frauwallner A (2013) Exercise, intestinal barrier dysfunction and probiotic supplementation. Med Sport Sci. https://doi.org/10.1159/000342169
21. Leite GSF, Resende Master Student AS, West NP, Lancha AH (2019) Probiotics and sports: A new magic bullet? Nutrition. https://doi.org/10.1016/j.nut.2018.09.023
22. Pyne DB, West NP, Cox AJ, Cripps AW (2015) Probiotics supplementation for athletes—clinical and physiological effects. Eur J Sport Sci. https://doi.org/10.1080/17461391.2014.971879
23. Adikari AMGCP, Adikari AMGCP, Appukutty M, Kuan G. Effects of Daily Probiotic Supplementation on Football Player's Stress and Anxiety. 2019 https://doi.org/10.2991/acpes-19.2019.1.
24. Wosinska L, Cotter PD, O'Sullivan O, Guinane C (2019) The potential impact of probiotics on the gut microbiome of athletes. Nutrients. https://doi.org/10.3390/nu11102270
25. Pugh JN, Sparks AS, Doran DA, Fleming SC, Langan-Evans C, Kirk B et al (2019) Four weeks of probiotic supplementation reduces GI symptoms during a marathon race. Eur J Appl Physiol. https://doi.org/10.1007/s00421-019-04136-3
26. Díaz-Jiménez J, Sánchez-Sánchez E, Ordoñez FJ, Rosety I, Díaz AJ, Rosety-Rodriguez M et al (2021) Impact of probiotics on the performance of endurance athletes: a systematic review. Int J Environ Res Public Health. https://doi.org/10.3390/ijerph182111576
27. Heaney S, O'Connor H, Michael S, Gifford J, Naughton G (2011) Nutrition knowledge in athletes: a systematic review. Int J Sport Nutr Exerc Metab. https://doi.org/10.1123/ijsnem.21.3.248
28. Spronk I, Kullen C, Burdon C, O'Connor H (2014) Relationship between nutrition knowledge and dietary intake. Br J Nutr. https://doi.org/10.1017/S0007114514000087
29. Rossi FE, Landreth A, Beam S, Jones T, Norton L, Cholewa JM. The effects of a sports nutrition education intervention on nutritional status, sport nutrition knowledge, body composition, and performance during off season training in NCAA division I baseball players. J Sports Sci Med 2017; 16.
30. Renard M, Kelly DT, Chéilleachair NN, Catháin C (2020) Evaluation of nutrition knowledge in female gaelic games players. Sports. https://doi.org/10.3390/sports8120154
31. Tam R, Beck KL, Manore MM, Gifford J, Flood VM, O'Connor H (2019) Effectiveness of education interventions designed to improve nutrition knowledge in athletes: a systematic review. Sports Med. https://doi.org/10.1007/s40279-019-01157-y
32. Trakman GL, Forsyth A, Hoye R, Belski R (2017) The nutrition for sport knowledge questionnaire (NSKQ): development and validation using classical test theory and Rasch analysis. J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-017-0182-y
33. Trakman GL, Brown F, Forsyth A, Belski R (2019) Modifications to the nutrition for sport knowledge questionnaire (NSQK) and abridged nutrition for sport knowledge questionnaire (ANSKQ). J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-019-0293-8
34. Trakman GL, Forsyth A, Devlin BL, Belski R (2016) A systematic review of athletes' and coaches' nutrition knowledge and reflections on the quality of current nutrition knowledge measures. Nutrients. https://doi.org/10.3390/nu8090570
35. Spendlove JK, Heaney SE, Gifford JA, Prvan T, Denyer GS, O'Connor HT (2012) Evaluation of general nutrition knowledge in elite Australian athletes. Br J Nutr. https://doi.org/10.1017/S0007114511005125
36. Laddu D, Hauser M (2019) Addressing the nutritional phenotype through personalized nutrition for chronic disease prevention and management. Prog Cardiovasc Dis. https://doi.org/10.1016/j.pcad.2018.12.004
37. Stover PJ, James WPT, Krook A, Garza C (2018) Emerging concepts on the role of epigenetics in the relationships between nutrition and health. J Intern Med. https://doi.org/10.1111/joim.12768
38. Yang Y, He D, Wei L, Wang S, Chen L, Luo M et al (2020) Association between diet-related knowledge, attitudes, behaviors, and self-rated health in Chinese adult residents: a population-based study. BMC Public Health. https://doi.org/10.1186/s12889-020-08896-y
39. Di Dio M, Calella P, Pelullo CP, Liguori F, Di Onofrio V, Gallè F et al (2023) Effects of probiotic supplementation on sports performance and performance-related features in athletes: a systematic review. Int J Environ Res Public Health. https://doi.org/10.3390/ijerph20032226
40. Di Dio M, Calella P, Cerullo G, Pelullo CP, Di Onofrio V, Gallè F et al (2022) Effects of probiotics supplementation on risk and severity of infections in athletes: a systematic review. Int J Environ Res Public Health. https://doi.org/10.3390/ijerph191811534
41. Soni R, Tank K, Jain N (2018) Knowledge, attitude and practice of health professionals about probiotic use in Ahmedabad, India. Nutr Food Sci. https://doi.org/10.1108/NFS-02-2017-0032
42. Fijan S, Frauwallner A, Varga L, Langerholc T, Rogelj I, Lorber M et al (2019) Health professionals' knowledge of probiotics: an international survey. Int J Environ Res Public Health. https://doi.org/10.3390/ijerph16173128
43. Jagim AR, Fields JB, Magee M, Kerksick C, Luedke J, Erickson J et al (2021) The influence of sport nutrition knowledge on body composition and perceptions of dietary requirements in collegiate athletes. Nutrients. https://doi.org/10.3390/nu13072239
44. Klein DJ, Eck KM, Walker AJ, Pellegrino JK, Freidenreich DJ (2021) Assessment of sport nutrition knowledge, dietary practices, and sources of nutrition information in ncaa division iii collegiate athletes. Nutrients. https://doi.org/10.3390/nu13092962
45. Torres-McGehee TM, Pritchett KL, Zippel D, Minton DM, Cellamare A, Sibilia M (2012) Sports nutrition knowledge among collegiate athletes, coaches, athletic trainers, and strength and conditioning specialists. J Athl Train. https://doi.org/10.4085/1062-6050-47.2.205

46. Tam R, Gifford JA, Beck KL (2022) Recent developments in the assessment of nutrition knowledge in athletes. Curr Nutr Rep. https://doi.org/10.1007/s13668-022-00397-1
47. Pentz R (2021) ES13.05 The Ethics of International Research. J Thoracic Oncol. https://doi.org/10.1016/j.jtho.2021.08.748
48. O'brien L, Collins K, Amirabdollhian F (2021) Exploring sports nutrition knowledge in elite gaelic footballers. Nutrients. https://doi.org/10.3390/nu13041081
49. Qualtrics. Qualtrics XM. Qualtrics. 2023.
50. Chapman SJ (2018) Review of discovering statistics using IBM SPSS statistics, 4th Edition. J Political Sci Educ. https://doi.org/10.1080/15512169.2017.1366328
51. Faul F, Erdfelder E, Lang AG, Buchner A. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. In: *Behavior Research Methods*. 2007. https://doi.org/10.3758/BF03193146.
52. Field A. Discovering statistic using IBM SPSS statistic 5th. *Dk* 2017; 53.
53. Han M, Yang K, Yang P, Zhong C, Chen C, Wang S et al (2020) Stratification of athletes' gut microbiota: the multifaceted hubs associated with dietary factors, physical characteristics and performance. Gut Microbes. https://doi.org/10.1080/19490976.2020.1842991
54. Stuempfle KJ, Hoffman MD, Hew-Butler T (2013) Association of gastrointestinal distress in ultramarathoners with race diet. Int J Sport Nutr Exerc Metab. https://doi.org/10.1123/ijsnem.23.2.103
55. Ten Haaf DSM, Van Der Worp MP, Groenewoud HMM, Leij-Halfwerk S, Nijhuis-van Der Sanden MWG, Verbeek ALM et al (2014) Nutritional indicators for gastrointestinal symptoms in female runners: the "Marikenloop study." BMJ Open. https://doi.org/10.1136/bmjopen-2014-005780
56. Parnell JA, Wagner-Jones K, Madden RF, Erdman KA (2020) Dietary restrictions in endurance runners to mitigate exercise-induced gastrointestinal symptoms. J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-020-00361-w
57. Stuempfle KJ, Hoffman MD (2015) Gastrointestinal distress is common during a 161-km ultramarathon. J Sports Sci. https://doi.org/10.1080/02640414.2015.1012104
58. Stuempfle KJ, Valentino T, Hew-Butler T, Hecht FM, Hoffman MD (2016) Nausea is associated with endotoxemia during a 161-km ultramarathon. J Sports Sci. https://doi.org/10.1080/02640414.2015.1130238
59. Ribichini E, Scalese G, Cesarini A, Mocci C, Pallotta N, Severi C et al (2023) Exercise-induced gastrointestinal symptoms in endurance sports: a review of pathophysiology, symptoms, and nutritional management. Dietetics. https://doi.org/10.3390/dietetics2030021
60. Urwin CS, Main LC, Mikocka-Walus A, Skvarc DR, Roberts SSH, Condo D et al (2021) The relationship between psychological stress and anxiety with gastrointestinal symptoms before and during a 56 km ultramarathon running race. Sports Med Open. https://doi.org/10.1186/s40798-021-00389-5
61. Diduch BK (2017) Gastrointestinal conditions in the female athlete. Clin Sports Med. https://doi.org/10.1016/j.csm.2017.06.001
62. Huang WC, Wei CC, Huang CC, Chen WL, Huang HY (2019) The beneficial effects of Lactobacillus plantarum PS128 on high-intensity, exercise-induced oxidative stress, inflammation, and performance in triathletes. Nutrients. https://doi.org/10.3390/nu11020353
63. Shing CM, Peake JM, Lim CL, Briskey D, Walsh NP, Fortes MB et al (2014) Effects of probiotics supplementation on gastrointestinal permeability, inflammation and exercise performance in the heat. Eur J Appl Physiol. https://doi.org/10.1007/s00421-013-2748-y
64. Cronin O, Molloy MG, Shanahan F (2016) Exercise, fitness, and the gut. Curr Opin Gastroenterol. https://doi.org/10.1097/MOG.0000000000000240
65. Mach N, Fuster-Botella D (2017) Endurance exercise and gut microbiota: a review. J Sport Health Sci. https://doi.org/10.1016/j.jshs.2016.05.001
66. Chantler S, Griffiths A, Matu J, Davison G, Jones B, Holliday A (2022) A systematic review: role of dietary supplements on markers of exercise-associated gut damage and permeability. PLoS ONE. https://doi.org/10.1371/journal.pone.0266379
67. Werner EN, Guadagni AJ, Pivarnik JM (2022) Assessment of nutrition knowledge in division I college athletes. J Am Coll Health. https://doi.org/10.1080/07448481.2020.1740234
68. Mitchell L, Lehane P, McCarthy M, O'Shea K, Tracey M, Whyte T et al (2022) Nutrition knowledge of elite and non-elite Gaelic footballers. Sci Med Football. https://doi.org/10.1080/24733938.2021.1906440
69. Renard M, Anton-Solanas A, Kelly DT, Ó Catháin C (2022) Evaluation of nutrition knowledge in elite and sub-elite Gaelic football players. Sci Med Football. https://doi.org/10.1080/24733938.2021.1883195
70. Ribeiro FM, Petriz B, Marques G, Kamilla LH, Franco OL (2021) Is there an exercise-intensity threshold capable of avoiding the leaky gut? Front Nutr. https://doi.org/10.3389/fnut.2021.627289
71. Nascimento M, Silva D, Ribeiro S, Nunes M, Almeida M, Mendes-Netto R (2016) Effect of a nutritional intervention in athlete's body composition, eating behaviour and nutritional knowledge: a comparison between adults and adolescents. Nutrients. https://doi.org/10.3390/nu8090535
72. Worsley A (2002) Nutrition knowledge and food consumption: Can nutrition knowledge change food behaviour? Asia Pac J Clin Nutr 11(Suppl):3. https://doi.org/10.1046/j.1440-6047.11.supp3.7.x
73. Nikolaidis PT, Veniamakis E, Rosemann T, Knechtle B (2018) Nutrition in ultra-endurance: state of the art. Nutrients. https://doi.org/10.3390/nu10121995
74. Morici G, Gruttad'Auria CI, Baiamonte P, Mazzuca E, Castrogiovanni A, Bonsignore MR (2016) Endurance training: Is it bad for you? Breathe. https://doi.org/10.1183/20734735.007016
75. Ter Steege RWF, Van Der Palen J, Kolkman JJ (2008) Prevalence of gastrointestinal complaints in runners competing in a long-distance run: an internet-based observational study in 1281 subjects. Scand J Gastroenterol. https://doi.org/10.1080/00365520802321170
76. Quero CD, Manonelles P, Fernández M, Abellán-Aynés O, López-Plaza D, Andreu-Caravaca L et al (2021) Differential health effects on inflammatory, immunological and stress parameters in professional soccer players and sedentary individuals after consuming a symbiotic. A triple-blinded, randomized, placebo-controlled pilot study. Nutrients. https://doi.org/10.3390/nu13041321
77. Huang WC, Pan CH, Wei CC, Huang HY (2020) Lactobacillus plantarum ps128 improves physiological adaptation and performance in triathletes through gut microbiota modulation. Nutrients. https://doi.org/10.3390/nu12082315
78. Cox AJ, Pyne DB, Saunders PU, Fricker PA (2010) Oral administration of the probiotic Lactobacillus fermentum VRI-003 and mucosal immunity in endurance athletes. Br J Sports Med. https://doi.org/10.1136/bjsm.2007.044628
79. Schreiber C, Tamir S, Golan R, Weinstein A, Weinstein Y (2021) The effect of probiotic supplementation on performance, inflammatory markers and gastro-intestinal symptoms in elite road cyclists. J Int Soc Sports Nutr. https://doi.org/10.1186/s12970-021-00432-6

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